Federated Learning for Mammography Screening Across Hospitals

(2024)

Problem

Training breast cancer detection models across multiple hospital sites without sharing raw mammography images.

Method

A federated learning framework using FedAvg combined with differential privacy noise injection.

Dataset

Multi-site private mammograms plus CBIS-DDSM for external validation.

Evaluation

Sensitivity and specificity at a fixed false-positive rate.

Results

The federated model reached 88% sensitivity at a fixed 10% false-positive rate, within 2 points of a centrally-trained model.

Limitations

Communication overhead scales poorly beyond 10 participating sites, limiting real-world hospital network deployment.